Interactive map of the archaeological heritage of Águeda

English | Português

An interactive web map of the 17 archaeological sites in the municipality of Águeda, Portugal, built from open data published by the municipal council. The data was processed in **PostgreSQL/PostGIS**, worked on in **QGIS**, and published with **qgis2web** on **GitHub Pages**.

This project continues the previous project, where the same data was imported and analysed with SQL and visualised in Power BI.

Result

Open the map: https://mykytalarionov603.github.io/mapa-interativo-patrimonio-agueda/

Clicking a point opens a box with the site name, parish, chronology, period, a description, the source and a link to more information. The legend shows the periods, and points are coloured by period.

Data

Dataset: "Património Arqueológico" (Archaeological Heritage), Águeda Municipal Council, on the Open Data Portal: https://dados.gov.pt/en/datasets/patrimonio-arqueologico-1/

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Origin: according to the portal, the archaeological heritage layer is part of the Heritage Zoning Map (Planta de Ordenamento – Património) of the Municipal Master Plan (PDM) in force (Diário da República, 2nd series, no. 44, 1 March 2012, Notice no. 3341/2012).

Content: 17 archaeological sites, in 17 columns: id, gid, codigo, dominio, subdominio, familia, objecto, ident_gene, ident_part, regula, cronologia, obs, freguesia, lat, lon, id_patri, id_orden.

Coordinates: in the CSV, in degrees, minutes and seconds (for example 40º 39' 12.41" N).

Site descriptions: taken from Câmara Municipal de Águeda, *Plano Director Municipal de Águeda – Revisão. Estudos Sectoriais: História e Património* (Águeda Municipal Master Plan – Revision, Sectoral Studies: History and Heritage).

Changes made to the data

1. Coordinates converted to decimal degrees.
2. Geometry column geom created.
3. Columns descricao, fonte and link added.
4. Chronology grouped into periods, in a derived column. **The original chronology text was not changed.**

The data is in Portuguese, so column names and period labels in the SQL below (for example Pré-história, Romano, Medieval) are kept as they appear in the project.

Tools

Tool	What it was used for
PostgreSQL 18 and pgAdmin 4	Storing, processing and querying the data
PostGIS	Creating the geometry (point) column
QGIS	Cartography: colours by period, labels, basemap
qgis2web (OpenLayers)	Exporting the project as a web map
GitHub Pages	Publishing the map at a public address
Esri World Street Map	Basemap

Starting point

The pa_test table already existed in the Pat ARQ database, imported in the previous project. The coordinates were converted from degrees/minutes/seconds to decimal degrees:

```sql
ALTER TABLE pa_test ADD COLUMN lat_dec numeric, ADD COLUMN lon_dec numeric;

UPDATE pa_test SET
  lat_dec =  (regexp_split_to_array(lat, '[^0-9.]+'))[1]::numeric
          + (regexp_split_to_array(lat, '[^0-9.]+'))[2]::numeric / 60
          + (regexp_split_to_array(lat, '[^0-9.]+'))[3]::numeric / 3600,
  lon_dec = -( (regexp_split_to_array(lon, '[^0-9.]+'))[1]::numeric
          + (regexp_split_to_array(lon, '[^0-9.]+'))[2]::numeric / 60
          + (regexp_split_to_array(lon, '[^0-9.]+'))[3]::numeric / 3600 );
```

Process, step by step

1. Connecting QGIS to PostgreSQL

In QGIS, **Browser → PostgreSQL → New Connection**, with these values:

Field	Value
Service	empty
Host	localhost
Port	5432
Database	Pat ARQ
Authentication	**Basic** tab, with username and password saved

2. Enabling PostGIS and creating the geometry

```sql
CREATE EXTENSION IF NOT EXISTS postgis;
```

```sql
ALTER TABLE pa_test ADD COLUMN geom geometry(Point, 4326);
```

```sql
UPDATE pa_test
SET geom = ST_SetSRID(ST_MakePoint(lon_dec, lat_dec), 4326);
```

Code 4326 is the WGS 84 coordinate system (the one GPS uses). After this, QGIS recognises pa_test as a point layer and the 17 sites appear on the map.

3. Basemap

Added in **Browser → XYZ Tiles → New Connection**:

```
https://server.arcgisonline.com/ArcGIS/rest/services/World_Street_Map/MapServer/tile/{z}/{y}/{x}
```

The earlier attempts with other basemaps are described in the challenges section.

4. Colours by period

The cronologia field is free text, with 13 different wordings ("Alta Idade Media", "Romano (?)", "Tardo romano/medieval", etc.). To keep the map readable, they were grouped into 6 categories with a CASE expression, **without changing the original text**:

```sql
CASE
  WHEN cronologia ILIKE '%milenio%' OR cronologia ILIKE '%pre-hist%' THEN 'Pré-história'
  WHEN cronologia ILIKE '%roman%' AND cronologia ILIKE '%medi%' THEN 'Várias épocas'
  WHEN cronologia ILIKE '%roman%' THEN 'Romano'
  WHEN cronologia ILIKE '%medi%' THEN 'Medieval'
  WHEN cronologia ILIKE '%proto%' THEN 'Proto-história'
  ELSE 'Indeterminada'
END
```

Period	Sites
Prehistory (Pré-história)	5
Protohistory (Proto-história)	1

Roman (Romano)	3
Medieval (Medieval)	3
Several periods (Várias épocas)	3
Undetermined (Indeterminada)	2

In QGIS, the symbology is **Categorized** by this field. The four ordered periods use a sequential colour ramp (lightest to darkest, in chronological order), while "Several periods" and "Undetermined" have colours outside that sequence.

Sites whose chronology is uncertain in the source (marked "(?)" or "Provavelmente", meaning "probably") keep the colour of the period indicated. The uncertainty remains visible in the original chronology text, which appears in the pop-up.

5. Labels

Full names are long, so the label shows only the part before the first comma, with two names shortened by hand:

```sql
CASE
  WHEN id = 17 THEN 'Torre de Menagem (?)'
  WHEN id = 5 THEN 'Povoado de Vila Verde'
  WHEN strpos(ident_part, ',') > 0 THEN left(ident_part, strpos(ident_part, ',') - 1)
  ELSE ident_part
END
```

The full name stays in the table and appears in the pop-up. Label text has a white outline so it reads clearly over the map.

6. Adding description, source and link

Three new columns were created:

```sql
ALTER TABLE pa_test ADD COLUMN descricao text;
ALTER TABLE pa_test ADD COLUMN fonte text;
ALTER TABLE pa_test ADD COLUMN link text;
```

And filled with a single UPDATE, where each line of VALUES carries the site id and its text (the id links each text to the right row):

```sql
UPDATE pa_test AS t
SET descricao = v.descricao, fonte = v.fonte, link = v.link
FROM (VALUES
  (1, 'text for site 1', 'source for site 1', 'https://...'),
```

```
  (2, 'text for site 2', 'source for site 2', 'https://...')
  -- ... up to site 17
) AS v(id, descricao, fonte, link)
WHERE t.id = v.id;
```

7. Final view for the pop-up

The qgis2web pop-up shows fields **in the order of the table columns** and with the column names. Since QGIS cannot reorder the columns of a PostgreSQL table, a *view* was created with the columns in the order and with the names to be displayed:

```
SQL
CREATE VIEW v_mapa_final AS
SELECT
  id,
  ident_part AS "Nome",
  freguesia AS "Freguesia",
  cronologia AS "Cronologia",
  CASE
    WHEN cronologia ILIKE '%milenio%' OR cronologia ILIKE '%pre-hist%' THEN 'Pré-história'
    WHEN cronologia ILIKE '%roman%' AND cronologia ILIKE '%medi%' THEN 'Várias épocas'
    WHEN cronologia ILIKE '%roman%' THEN 'Romano'
    WHEN cronologia ILIKE '%medi%' THEN 'Medieval'
    WHEN cronologia ILIKE '%proto%' THEN 'Proto-história'
    ELSE 'Indeterminada'
  END AS "Época",
  descricao AS "Descrição",
  fonte AS "Fonte",
  link AS "Link",
  CASE
    WHEN cronologia ILIKE '%milenio%' OR cronologia ILIKE '%pre-hist%' THEN 'Pré-história'
    WHEN cronologia ILIKE '%roman%' AND cronologia ILIKE '%medi%' THEN 'Várias épocas'
    WHEN cronologia ILIKE '%roman%' THEN 'Romano'
    WHEN cronologia ILIKE '%medi%' THEN 'Medieval'
    WHEN cronologia ILIKE '%proto%' THEN 'Proto-história'
    ELSE 'Indeterminada'
  END AS epoca,
  CASE
    WHEN id = 17 THEN 'Torre de Menagem (?)'
    WHEN id = 5 THEN 'Povoado de Vila Verde'
    WHEN strpos(ident_part, ',') > 0 THEN left(ident_part, strpos(ident_part, ',') - 1)
    ELSE ident_part
  END AS rotulo,
  geom
FROM pa_test;
```

The period appears twice: "Época" is the one shown in the pop-up, and epoca is a copy, hidden in the pop-up, used for the colours. The rotulo column is used for the labels.

In QGIS, the view was loaded as a layer, the style of the old layer was copied (**Layer → Copy Style / Paste Style**) and the layer was renamed "Património arqueológico", which becomes the pop-up title.

8. Exporting with qgis2web

Web → qgis2web → Create web map, **OpenLayers** library:

Option	Value
Visible layers	only "Património arqueológico" and the Esri basemap
Pop-up fields	Nome as the title; Freguesia, Cronologia, Época and Fonte inline; Descrição and Link with no label; id, epoca, rotulo and geom hidden
Show popups on hover	unchecked (the pop-up opens on click)
Highlight on hover	checked
Layers list	Expanded

9. Publishing on GitHub Pages

1. Create a public repository.
2. Drag the exported folder (with index.html and its subfolders) into **Add file → Upload files**.
3. In **Settings → Pages**, choose **Deploy from a branch**, the main branch and the / (root) folder.
4. Wait a minute or two. The map address appears in **Settings → Pages**.

Challenges and how I solved them

Problem	Cause	Solution
QGIS: definition of service "localhost" not found	The **Service** field was filled in with "localhost"	Leave Service empty and enter localhost in **Host**
Empty public schema in QGIS	The connection pointed to the postgres database, while the table is in Pat ARQ	Correct the **Database** field
extension "postgis" is not available	PostGIS was not installed on the system	Install PostGIS with PostgreSQL's Stack Builder and repeat CREATE EXTENSION
OpenStreetMap basemap showing "Access blocked" (error 403) on the exported page	OpenStreetMap refuses requests that do not identify their origin, and a page opened from disk does not send it	Use another basemap (Esri World Street Map)
CARTO basemap showing "API KEY REQUIRED"	CARTO now requires an API key	Stop using this basemap

"Mapzen Global Terrain" basemap dark and unreadable	It appears to be a colour-coded elevation map rather than a map meant to be viewed	Switch to Esri World Street Map
qgis2web raised a Python error (TypeError ... argument 1 has unexpected type 'list')	The plugin cannot translate CASE expressions used in labels into JavaScript	Move the logic into fields (and later into the *view*), instead of expressions
Empty labels and all-identical colours after creating virtual fields	The fields had been created with the **number** type instead of **text**	Create the fields with the **Text (string)** type
Pop-up fields in the wrong order and showing the table's column names	The plugin follows the order and names of the columns	Create the v_mapa_final *view*
The pop-up box disappeared when moving the mouse to the link	In "on hover" mode the box closes as soon as the mouse leaves the point	Turn off "Show popups on hover": the pop-up opens on click and stays open
GitHub only accepted index.html and the README when uploading files	The file picker does not accept folders	Drag the whole folder into the upload area
The GitHub Pages address did not appear	Pages does not switch on by itself	Enable it in **Settings → Pages**

Decisions and limitations

The original chronology text was never changed. The grouping by period is used only for colours, and the original chronology appears in each site's pop-up.

The grouping is a simplification. The chronology is free text, and the classification was done with keywords. It should be reviewed by someone who knows the chronology of the sites.

Uncertain chronologies ("(?)", "Provavelmente") keep the colour of the period indicated.

The 17 sites are the inventoried ones. The map shows where known sites are, which also depends on where surveys were carried out, and it does not allow conclusions about the real distribution of ancient settlement.

Coordinate system. The portal refers to PT-TM06/ETRS89 for the geographic information, and the CSV coordinates were treated as WGS 84 (EPSG:4326). The difference between the two is small (of the order of a metre) and does not affect how the map reads.

Descriptions. Each pop-up states the source of the text.

Reusing the map

The map is a static site (HTML and JavaScript), so it needs neither a database nor QGIS to work.

Open the link of the published map.

Embed it in another site with an iframe:

```HTML
<iframe src="https://mykytalarionov603.github.io/mapa-interativo-patrimonio-agueda/"
        width="100%" height="600" style="border:0"
        title="Archaeological heritage of Águeda"></iframe>
```

Download the map folder (**Code → Download ZIP** button) and host it on another server.

The basemap is a third-party service. According to Esri's service description, World Street Map is in "mature support" and is no longer updated, and the terms of use may change. For institutional or long-term use, it is worth confirming them and, if necessary, switching to a licensed basemap.

Skills demonstrated

Connecting QGIS to a PostgreSQL/PostGIS database

SQL: ALTER, UPDATE ... FROM (VALUES ...), CASE, *views* and geospatial functions (ST_MakePoint, ST_SetSRID)

Thematic cartography: categorized symbology, labels and legend

Publishing an interactive web map

Troubleshooting and documentation

Credits and licences

Data: © Câmara Municipal de Águeda, "Património Arqueológico", licence CC BY 4.0, https://dados.gov.pt/en/datasets/patrimonio-arqueologico-1/. The data was modified (see "Changes made to the data").

Site descriptions: Câmara Municipal de Águeda, *Plano Director Municipal de Águeda – Revisão. Estudos Sectoriais: História e Património*. Information about the PDM: https://www.cm-agueda.pt/viver/urbanismo-ordenamento-do-territorio-e-politicas-de-habitacao/planos-municipais-de-ordenamento-territorio/plano-diretor-municipal-de-agueda

Basemap: Esri World Street Map. Sources: Esri, HERE, Garmin, USGS, Intermap, INCREMENT P, NRCan, Esri Japan, METI, Mapwithyou, NOSTRA, © OpenStreetMap contributors, and the GIS user community. Terms of use: https://goto.arcgisonline.com/maps/World_Street_Map

Software: QGIS, PostgreSQL, PostGIS, qgis2web plugin.